

Top-Tier Manuscript Architecture v1

Policy-Level Search Size for Adaptive Statistical Pipelines

Strategic status: Post-D2 integration review

Date: 2 August 2026

Theory branch: tess-top-tier-theory

Current theory head: 38daf14 (tess-theory-work-package-d2-v1-results-final-20260802)

Fallback empirical branch: ess-study

Purpose: Decide the scientific identity, theorem spine, journal route, manuscript structure, and minimum remaining work before drafting a top-tier submission.

Decision: one theory-first paper; Biometrika first; D4 before D3

1. Executive decision

1.1 Overall judgment

The project has crossed the threshold from a strong applied-methodology paper to a credible foundational statistics paper.

The reason is **not** the logarithmic TESS formula by itself. The viable foundational claim is:

Fixed candidate laws and expected candidate-count budgets do not determine the inferential burden of an adaptive policy. The missing object is the threshold-indexed coupling between the policy's branching rule and its counterfactual incremental rejection opportunities.

Work Packages A and B supply the structural theory. Work Packages D1 and D2 supply regular two-bank inference. The locked confirmatory study supplies a nontrivial empirical phenomenon that the theory explains rather than merely illustrates.

1.2 Recommended publication strategy

1. **Prepare one theory-first manuscript.** It should supersede, not merely append to, the current empirical v2 draft.
2. **Retain empirical v2 as a frozen fallback.** Do not submit the theory paper and the current empirical TESS paper as overlapping separate publications.
3. **First target: Biometrika, regular paper,** provided the main paper can remain under 20 pages and the theorem scope is tightened as specified below.
4. **Second target: JRSS-B,** if the paper needs more space for methodology, application scope, and the locked empirical illustration.
5. **JASA Theory and Methods** becomes especially attractive if a uniform process theorem, software implementation, or a broader inference package is added.
6. **Annals of Statistics is not the immediate target.** It becomes realistic only if a substantially more general process theory or policy-class result becomes the central contribution.

1.3 Next theorem decision

Develop D4, the paired budget-matched policy-contrast theorem, before D3.

D4 directly connects the sharp allocation theory to the locked promising-minus-matched-random estimand. It is the smallest missing theorem that closes the conceptual and inferential loop of the paper.

D3, a uniform TESS-process theorem and simultaneous bands, remains valuable but should be attempted only after a complete manuscript draft shows that it is necessary for the target journal.

2. Scientific identity of the paper

2.1 Recommended title

Preferred

Inferential Search Size of Adaptive Statistical Policies

Strong alternative

Search Size Is a Property of the Adaptive Policy

More descriptive alternative

Policy-Level Search Size under Adaptive Statistical Branching

Avoid putting “TESS” in the title. The acronym is useful inside the paper, but the foundational object is policy-level search size, not a branded formula.

2.2 One-sentence paper claim

Two adaptive policies can share the complete fixed candidate law, marginal inclusion probabilities, candidate-count distribution, and expected candidate cost, yet have different global-null rejection curves; policy-level search size is determined by how activation is coupled to threshold-specific incremental rejection.

2.3 Abstract-level contribution statement

The final abstract should communicate four layers, in this order:

1. **Non-identifiability:** fixed-family dependence and budget summaries are insufficient.
2. **Sharp policy calculus:** the policy rejection law is generated by the coupling between activation and conditional incremental rejection; sharp same-budget envelopes follow.
3. **Threshold structure:** a common activation ordering is uniformly optimal only under rank coherence; otherwise policy curves can reverse.
4. **Inference and evidence:** reference- and evaluation-bank uncertainty admit a two-bank expansion and bootstrap; a locked machine-learning experiment demonstrates the phenomenon.

2.4 What the paper is not

- not the first effective-number-of-tests formula;
- not a new extremal coefficient;
- not a universal multiplicity correction;
- not a claim that every adaptive procedure is harmful;
- not a replacement for exact pipeline-aware max-statistic inference;
- not primarily a clinical prediction paper;
- not a paper whose novelty rests on a covariance identity alone.

3. Source and evidence hierarchy

3.1 Frozen theoretical sources

Component	Frozen source	Status in final paper
Axiomatic characterization and fixed-family copula bridge	Theory memo v1	Short supporting proposition/remark
Sharp policy representation and envelope	Work Package A	Central theorem
Identifiability and fixed-budget impossibility	Work Package A	Central corollaries/theorem
Cost-weighted allocation	Work Package A	Supplement
Threshold-rank coherence	Work Package B	Main theorem after proof audit
Full-curve stochastic dominance	Work Package B	Supplement or corollary
Arbitrary policy-curve crossings	Work Package B	Main counterexample after scope repair
Known-trigger two-bank inference	D1	Supplementary stepping stone; REVIEW retained
Estimated-trigger two-bank inference	D2	Main inference theorem; PASS
Conditional DKW bands	Theory memo v1	Supplementary rigorous baseline

3.2 Frozen empirical sources

Evidence	Role
Fresh-seed locked confirmatory study	Main empirical illustration
Promising vs budget-matched random primary contrast	Main result
Rescue policy, high-dependency library, alpha curves	Secondary empirical structure
Reference-bank uncertainty sensitivity	Supports the need for two-bank inference
SUPPORT2-anchored permutation experiment	Supplementary structural validation
Reconstruction, tie, failed-fit, archive, and hash audits	Reproducibility supplement

3.3 Evidence hierarchy in the final paper

1. General theory and impossibility results.
2. Regular two-bank estimation and bootstrap theory.
3. Locked confirmatory empirical illustration.
4. SUPPORT2-anchored and other supplementary checks.

This is the reverse of the current empirical v2 narrative. The top-tier paper must be theory-first.

4. Theorem audit and maturity map

4.1 Green: ready for manuscript-level proof polishing

Result	Why it is ready	Main/supplement
Independent-composition characterization	Correct, concise Cauchy-equation argument; novelty modest but role clear	Main, short
Fixed-family copula/extremal-coefficient representation	Correct bridge to prior art	Main remark or supplement
Policy representation $\pi_a = \pi_0 + E(a, m_\alpha)$	Fundamental and transparent	Main
Sharp same-rate envelope	Standard rearrangement applied to a new policy functional	Main
Identifiability iff $m_\alpha(S)$ is constant	Sharp and central	Main
Fixed expected candidate count does not bound deep-tail search size	Striking impossibility result	Main
D2 two-bank linear expansion	Regular empirical-process/quantile theorem, numerically validated	Main
D2 complete-vector bootstrap validity	Direct practical consequence, numerically validated	Main

Result	Why it is ready	Main/supplement
Exact finite-reference expectation in the independent benchmark	Explains D1 finite-B behavior	Supplement

4.2 Amber: retain, but complete a formal proof/scope audit before final drafting

Result	Issue to resolve	Required action
Rank-coherence/common-rank theorem	Uncountable threshold index requires a common version, separability, or a finite/dense-index formulation	State on compact separable I, add measurability/version conditions, or first state for finite threshold sets
Arbitrary smooth curve-crossing theorem	Current construction is naturally a branch-output selection policy; it must not be silently described as nested candidate addition	Rephrase as general adaptive branch selection, or rebuild a nested-expansion construction
Multistage telescoping extension	Correct algebraically but peripheral and likely to invite scope expansion	Move to a remark/supplement
Weighted compromise objective	Useful but secondary	Supplement
Trigger-only variance diagnostic	Ratio is unstable when the theoretical denominator is near zero	Report absolute contribution, not ratio, in any final table

4.3 Red: missing bridge results required before submission

1. **D4 paired policy-contrast theorem.**
2. **Exact plus-one/order-statistic bridge.**
3. **Finite-candidate unique-winner extension or an explicit statement that the inference theorem is a regular surrogate model.**
4. **A final literature novelty audit specifically targeting adaptive subset evaluation and selected-family search-cost functionals.**

5. Recommended theorem spine

The main paper should contain a small number of named results. Supporting algebra and secondary extensions move to the supplement.

Proposition 1. Characterization of the search-size scale

At fixed local threshold α , null-law invariance, continuity, independent-union additivity, and exact-single normalization uniquely give

$$S_A(\alpha) = \log\{1 - \pi_A(\alpha)\} / \log(1 - \alpha).$$

Role: justifies the scale.

Novelty claim: supporting, not central.

Remark 1. Fixed-family limit

For a fixed family of exact p-values, search size is the diagonal log-ratio of the survival copula; under extreme-value dependence it equals the extremal coefficient.

Role: locates the new object relative to known effective-number theory.

Placement: no more than half a page in the main text.

Theorem 1. Policy representation and sharp same-budget envelope

For activation-time information S , activation kernel $a(S)$, and incremental rejection D_α , define

$$m_\alpha(S) = E(D_\alpha \mid S).$$

Then

$$\pi_a(\alpha) = \pi_0(\alpha) + E\{a(S)m_\alpha(S)\},$$

and at fixed activation rate r , the attainable rejection probabilities are exactly the lower and upper integrated-quantile envelope of $m_\alpha(S)$.

Role: central representation theorem.

Corollary 1. Exact policy identifiability criterion

The fixed candidate law and activation rate identify policy rejection burden if and only if $m_\alpha(S)$ is almost surely constant.

Role: central conceptual claim.

Corollary 2. Budget insufficiency

Expected candidate count alone does not upper-bound deep-tail policy search size. At a fixed expected candidate-count budget, the deep-tail search size may diverge as the optional family grows.

Role: memorable impossibility result.

Theorem 2. Threshold coherence and simultaneous optimality

Under a formally audited rank-coherence condition, a common rank variable orders $m_\alpha(S)$ for all α in the declared interval; top-rank activation is simultaneously optimal throughout the interval.

Role: explains when one scalar trigger can be uniformly meaningful.

Theorem 3. Threshold-dependent non-orderability

Without rank coherence, policies with the same fixed branch law and the same budget can have crossing rejection and search-size curves. A smooth construction can place prescribed finite crossing points.

Role: curve-level impossibility result.

Scope warning: describe it as a general branch-output policy unless a nested-expansion construction is supplied.

Theorem 4. Estimated-trigger two-bank inference

Under the regular one-candidate-per-branch model, jointly estimating base, optional, and trigger quantiles from a complete-vector reference bank gives

$$\hat{\pi} - \pi = P_n \phi_E + P_B \phi_R + o_p(n^{-1/2} + B^{-1/2}),$$

with

$$\phi_R = a_0 W_0 + b_1 W_1 + d_U W_U,$$

and a valid independently resampled complete-replication two-bank bootstrap. The TESS result follows by the delta method.

Role: main inference theorem.

Theorem 5. Paired budget-matched policy contrast - D4, to be completed

This theorem should target the empirical mechanism directly.

Let

- $e_0 = E(R_0)$,
- $r = E(A)$,
- $\mu = E(D)$,
- $\nu = E(AD)$.

Then

- adaptive-policy rejection: $\pi_A = e_0 + \nu$,
- budget-matched random rejection: $\pi_C = e_0 + r \mu$,
- allocation premium: $\delta = \nu - r \mu = \text{Cov}(A, D)$,
- TESS contrast: $\Delta_S = g_\alpha(\pi_A) - g_\alpha(\pi_C)$.

The evaluation influence function has the explicit form

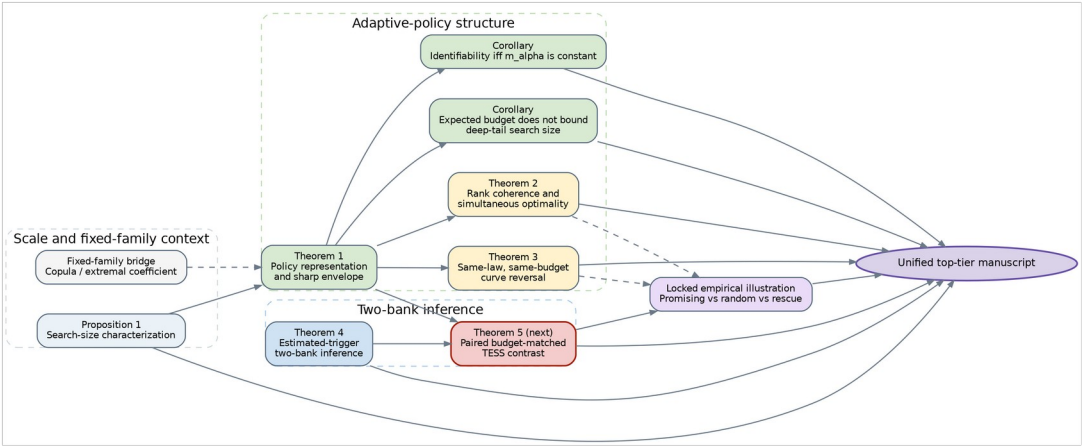
$$\phi_E^\Delta = g'(\pi_A)\{R_0 + AD - \pi_A\}$$

$$- g'(\pi_C)\{(R_0 - e_0) + \mu(A - r) + r(D - \mu)\}.$$

The reference contribution is the gradient of Δ_S with respect to the reference-calibrated policy parameters multiplied by their joint reference influence function. A paired complete-replication two-bank bootstrap should consistently estimate the contrast law.

Role: closes the loop between Work Package A, D2, and the locked primary estimand.

Theorem dependency map



Theorem and evidence dependency map

The red node is the single recommended new theorem before full manuscript drafting.

6. Why D4 should precede D3

Criterion	D4: paired policy contrast	D3: uniform process/bands
Direct match to locked primary estimand	Excellent	Indirect
Connects structural and inferential theory	Excellent	Moderate
Likely technical difficulty	Moderate	High
Required for Biometrika viability	Strongly desirable	No
Required for JRSS-B viability	Desirable	Helpful
Required for JASA-level inference package	Desirable	Strongly desirable
Risk of delaying manuscript indefinitely	Lower	Higher
Manuscript page cost	Low-moderate	High

Decision: develop and validate D4. Draft the complete paper. Reassess D3 only after the draft exists.

D3 should be added only if one of the following becomes true:

1. editors/reviewers are likely to view fixed-threshold inference as incomplete;
2. the paper is targeted to JASA with inference as a coequal contribution;
3. a concise process theorem follows with little additional notation;
4. simultaneous curve bands materially improve the empirical analysis.

7. Main-paper architecture for a Biometrika-first submission

7.1 Target length

Biometrika regular papers are normally fewer than 20 pages. The proposed main text should target **17-19 typeset pages including references, figures, and tables**, with full proofs and validation details in a separate supplement.

7.2 Section plan and page budget

Section	Purpose	Target pages
Abstract	Non-identifiability, sharp calculus, two-bank inference, locked illustration	0.4
1. Introduction	State the missing policy layer; distinguish fixed-family work	1.6-1.9
2. Search-size functional	Definition, characterization, fixed-family bridge	1.2-1.5
3. Adaptive policy calculus	Representation, envelope, identifiability, budget impossibility	3.0-3.4
4. Threshold ordering	Coherence, common trigger, crossing counterexample	2.2-2.6
5. Two-bank inference	D2 and D4 statements, interpretation, bootstrap	2.6-3.0
6. Locked illustration	One compact confirmatory example; no full methods dump	2.0-2.4
7. Discussion	Meaning, scope, error-control distinction, limitations	1.2-1.5
References	Core references only	2.0-2.5
Total		15.8-18.8

7.3 Detailed section outline

1. Introduction

Paragraph 1: adaptive statistical development creates state-dependent candidate sets.

Paragraph 2: fixed-family effective-number, data-snooping, post-selection, and adaptive-testing literatures do not identify the burden of a declared branching policy.

Paragraph 3: same budget does not imply same rejection law.

Paragraph 4: contributions, stated without introducing every theorem.

Paragraph 5: locked empirical illustration and paper map.

2. Policy-level search size

2.1 Policy rejection curve and exact/super-uniform distinction.

2.2 Independent-composition characterization.

2.3 Fixed-family copula/extremal-coefficient bridge and explicit prior-art boundary.

3. Adaptive policy calculus

3.1 Counterfactual base/full rejection and activation information.

3.2 Representation theorem.

3.3 Sharp same-rate envelope.

3.4 Identifiability criterion.

3.5 Fixed expected-budget impossibility result.

A one-paragraph cost-weighted remark may point to the supplement.

4. Threshold coherence and reversal

4.1 Rank coherence and common-index simultaneous optimality.

4.2 Why a score optimal at $\alpha = 0.05$ need not be optimal elsewhere.

4.3 One decisive curve-crossing construction.

Move FOSD, weighted compromise, multiple-crossing algebra, and regret to the supplement.

5. Two-bank estimation and policy contrasts

- 5.1 Population, conditional finite-reference, and unconditional two-bank targets.
- 5.2 D2 estimated-trigger linear expansion.
- 5.3 Complete-vector bootstrap.
- 5.4 D4 paired matched-budget contrast.
- 5.5 Finite-reference bias and exact plus-one bridge, summarized briefly.

6. Locked machine-learning illustration

- 6.1 Design in one compact paragraph.
 - 6.2 Primary promising-minus-random result.
 - 6.3 Rescue reversal and high- versus mixed-dependency comparison.
 - 6.4 Mechanism: signed gain/loss decomposition and threshold coherence diagnostic.
 - 6.5 Two-bank sensitivity.
- SUPPORT2 moves to the supplement, with one sentence in the main text.

7. Discussion

- policy multiplicity is a coupling property;
 - expected search budget is not an upper bound;
 - threshold ordering is not automatic;
 - TESS is diagnostic, while exact pipeline-aware inference controls error;
 - regular inference theorem versus exact winner-selection pipeline;
 - implications for reporting adaptive development policies;
 - limitations and extensions.
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8. Main figures and tables

Figure 1. The missing policy layer

A new conceptual figure, replacing the current workflow-only Figure 1.

Three columns:

1. **Fixed candidate law:** candidate scores/p-values and dependence.
2. **Adaptive policy layer:** activation information and kernel $a(S)$.
3. **Policy rejection law:** $\pi_a = \pi_0 + E(a \mid m_\alpha)$ and search-size curve.

The figure should visually show that candidate law + budget is insufficient without the coupling.

Figure 2. Sharp same-budget envelope and budget impossibility

Panel A: lower/random/upper attainable rejection or TESS at fixed activation rate.

Panel B: fixed expected candidate count with deep-tail search size increasing with optional-family size.

Figure 3. Coherence versus reversal

Panel A: common-index ordering (top > random > bottom) over the full curve.

Panel B: same-law, same-budget policies whose curves cross at a prescribed threshold.

Figure 4. Locked empirical illustration

A compact composite:

- Panel A: promising and rescue TESS contrasts in high-dependency, mixed-realistic, and SUPPORT2 structures.
- Panel B: signed mechanism contribution or $\text{Cov}(A, D_{\alpha})$ over α .

Existing detailed fixed-search curves, score-decile plots, and SUPPORT2 curves move to the supplement.

Table 1. Theorem map and assumptions

Columns: result, policy class, threshold scope, reference calibration, main assumptions, novelty role.

Table 2. Locked empirical results

Only the central contrasts, covariance, activation rate, gain/loss capture, and two-bank sensitivity.

9. Supplementary architecture

Supplement A. Complete proofs

- axiomatic characterization;
- sharp envelope and identifiability;
- unbounded budget mismatch;
- rank coherence and crossing results;
- cost-weighted allocation and stochastic dominance;
- D2 and D4 proof details.

Supplement B. Inference extensions

- D1 known-trigger derivation and REVIEW adjudication;
- D2 estimated-trigger derivation and PASS validation;
- exact finite-B expectation;
- conditional DKW bands;
- plus-one/order-statistic bridge;
- finite-candidate unique-winner extension;
- optional D3 process theorem if completed.

Supplement C. Numerical validation

- D1 and D2 locked protocols;

- DGP definitions;
- all cell summaries and diagnostics;
- explanation of basic interval undercoverage;
- trigger-only near-zero-denominator diagnostic.

Supplement D. Locked empirical study

- full simulation engine and candidate manifests;
- all policy definitions and tie rules;
- confirmatory and SUPPORT2 protocols;
- complete results tables;
- reconstruction and failed-fit audits;
- figure set currently in the empirical manuscript;
- raw archive and Git provenance.

Supplement E. Reproducibility

- software versions;
 - repository and DOI;
 - configuration SHA-256 values;
 - tags and commits;
 - AI assistance disclosure;
 - data and code availability.
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10. Mandatory bridge work before drafting the final manuscript

10.1 D4 paired contrast

Complete theory, code, locked numerical protocol, scientific validation, and results freeze.

10.2 Exact plus-one/order-statistic lemma

Derive the exact event equivalence for

$$p_{\text{hat}}_j(x) = \{1 + \sum I(T_{\text{bj}} \geq x)\} / (B+1)$$

under the strict rule $p_{\text{hat}}_j(x) < \alpha$, including:

- the integer rounding convention;
- continuous-score no-tie form;
- relation to an empirical upper quantile;
- the first-order equivalence regime;
- the grid/deep-tail regime in which discreteness remains first-order.

This bridge is necessary because the empirical application uses plus-one p-values, whereas D2 uses regular empirical quantiles.

10.3 Scope repair for the curve-crossing construction

Choose one of two honest routes:

1. **General branch-policy route:** state the theorem for a policy that selects between branch outputs; make clear that it proves non-orderability for general adaptive policies, not specifically nested candidate addition.
2. **Nested-expansion route:** construct the same crossing phenomenon with base/full counterfactual outputs arising from a nested candidate family.

Do not leave the scope implicit.

10.4 Proof audit of rank coherence

- specify whether I is finite, countable dense, or compact with separable versions;
- identify a common probability-one set;
- handle atoms through explicit boundary randomization;
- separate equivalence for all activation rates from statements at one rate.

10.5 Frozen-output empirical extraction

No model refitting is required. Extract and report:

1. loss counts $D_{\alpha} = -1$;
2. loss-capture fractions;
3. positive and negative covariance contributions;
4. exact mixed-minus-high interaction and CI;
5. exact SUPPORT2 rescue CI;
6. estimated $m_{\alpha}(S)$ on a common score-bin grid;
7. rank correlations of the estimated fields across α ;
8. FOSD diagnostics for promising/random/rescue activation scores;
9. observed curve crossings and simultaneous regret relative to the empirical pointwise envelope.

11. Overlap and publication-integrity guardrails

11.1 Relationship to the companion pipeline-aware inference paper

The distinction must be explicit in the introduction, discussion, cover letter, and repository README.

Companion pipeline-aware inference paper	Policy search-size paper
Provides valid search-adjusted max-statistic inference	Describes and estimates the naive policy-level global-null rejection burden
Primary object is an adjusted p-value/test	Primary object is a rejection-law functional and its search-size standardization

Companion pipeline-aware inference paper	Policy search-size paper
Re-executes the complete pipeline under the null Goal is error control	Studies how policy structure and calibration determine the burden Goal is measurement, structural understanding, and policy comparison

11.2 Relationship to empirical v2

The top-tier manuscript should absorb the key empirical evidence. The current empirical v2 remains a fallback, not a simultaneously submitted separate manuscript.

If the top-tier paper is abandoned, return to empirical v2. If the top-tier paper is submitted, do not submit empirical v2 unless it is later redesigned around a substantively distinct clinical/application question with nonoverlapping claims and figures.

12. Literature positioning and wording guardrails

12.1 Main prior-art clusters

- fixed-family effective numbers and Sidak/copula calibration;
- extremal coefficients and level-dependent tail dependence;
- data snooping and best-model bootstrap tests;
- search degrees of freedom;
- post-selection inference;
- adaptive data analysis;
- interactive, online, and learn-then-test procedures;
- selected-family and hierarchical multiple testing;
- empirical-process and two-sample bootstrap theory.

12.2 Safe novelty language

Use:

- “policy-level rejection functional”;
- “sharp same-budget envelope”;
- “policy non-identifiability from fixed-family and budget summaries”;
- “threshold-rank coherence”;
- “complete-replication two-bank inference”;
- “locked empirical illustration.”

Avoid:

- “the first effective number of tests”;
- “a new extremal coefficient”;
- “the first treatment of adaptive multiplicity”;
- “universal search complexity”;
- “TESS corrects multiplicity”;

- “D2 proves validity for the exact 20-candidate pipeline.”

12.3 Recommended narrow novelty paragraph

Fixed-family effective-number and extremal-coefficient methods describe the tail geometry of a declared family of statistics. They do not identify the rejection law of a policy whose evaluated family depends on intermediate data. We characterize the missing policy layer through conditional incremental rejection, derive sharp same-budget envelopes and identifiability conditions, show that identical fixed candidate laws and budgets permit different or crossing policy rejection curves, and develop two-bank inference when the policy calibration is estimated from an independent reference sample.

13. Reviewer-risk register

Risk	Severity	Response before submission
“TESS is just a Sidak inversion”	High	Make it a short characterized scale; center novelty on policy non-identifiability and envelopes
Curve-crossing theorem scope does not match nested expansion	High	Repair scope or construction before drafting
D2 is only a simplified two-candidate regular model	High	Add plus-one bridge, D4, and explicit limitation; never overclaim exact-pipeline validity
Too many theorems for one concise paper	High	Use the main theorem spine above; move cost/FOSD/D1/full validation to supplement
Empirical and theory papers appear duplicative	High	Submit one unified top-tier paper; freeze empirical v2 as fallback
Rank-coherence proof over uncountable alpha is informal	Medium-high	Add separability/version conditions or finite-grid theorem first
Expected-budget impossibility appears adversarial only	Medium	Pair theorem with locked heterogeneous-library phenomenon
D2 validation PASS depends on selecting $B \geq 3000$ as main	Medium	Explain this was prospectively informed by frozen D1 and locked before D2 run
Single-author proof reliability	Medium	Obtain independent mathematical review before submission
Manuscript exceeds Biometrika length	Medium	Enforce page budget; place proofs and SUPPORT2 in supplement
Novelty search misses adjacent adaptive-subset literature	Medium	Perform a focused systematic novelty audit and citation chaining

14. Journal route

14.1 Biometrika - recommended first submission

Best fit if the paper is concise and organized around:

- a foundational policy-level object;
- sharp non-identifiability/envelope results;
- unforeseen behavior under equal budgets;
- one compact inference theorem;
- one convincing locked illustration.

Minimum before submission: D4, plus-one bridge, scope repair, proof audit, and concise manuscript under 20 pages.

14.2 JRSS-B - recommended second route

Best fit if:

- the methodology and scope of application need more explanation;
- the locked empirical illustration remains substantial;
- D3 or broader computational implementation is included;
- the manuscript cannot honestly be compressed to Biometrika length.

A regular contributed paper is appropriate; it need not be a Discussion Paper.

14.3 JASA Theory and Methods

Best fit if:

- D3 supplies a uniform process and simultaneous bands;
- D4 is complete;
- software and a reusable policy-diagnostic workflow are part of the contribution;
- the paper is broader and more comprehensive than a concise Biometrika paper.

14.4 Annals of Statistics

Reconsider only if:

- inference is extended to a substantially more general policy class;
 - process weak convergence and bootstrap validity are central rather than supplementary;
 - winner selection or nonsmooth policy maps are handled nontrivially;
 - the theoretical advance is likely to influence statistical methodology beyond this application.
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15. Drafting sequence

1. Freeze this architecture and decision log.
2. Complete a proof audit of the green/amber theorem inventory.
3. Develop D4 under a separately locked protocol.
4. Derive and test the plus-one bridge.
5. Perform frozen-output signed-mechanism and coherence diagnostics.
6. Write a unified notation sheet and theorem dependency map.
7. Draft Sections 2-5 first.
8. Draft the empirical illustration from frozen outputs.
9. Draft Introduction and Discussion with the literature audit beside them.
10. Produce a Biometrika-length main PDF and a complete supplement.

11. Decide whether D3 is necessary only after the first complete typeset draft.
 12. Obtain external mathematical review, then conduct a final citation and overlap audit.
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16. Stop rules

Do not continue adding theory indefinitely.

Proceed without D3 if:

- D4 is complete;
- the plus-one bridge is satisfactory;
- the main theorem spine fits coherently under 20 pages;
- the locked illustration demonstrates the coupling mechanism;
- no major proof gap remains.

Attempt D3 only if it materially improves the paper's journal fit or closes an obvious inferential gap.

Return to empirical v2 if:

- the B-theorem scope cannot be repaired cleanly;
 - D4 requires assumptions unrelated to the empirical policy;
 - the unified main paper cannot be made concise;
 - a focused literature audit finds that the central policy-envelope/non-identifiability result is already known in essentially the same form.
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17. Immediate next action

The next work package should be:

Work Package D4: paired two-bank inference for the budget-matched policy TESS contrast.

Before any D4 scientific run:

1. write a D4 theory/protocol memo;
2. define the population target and its plug-in estimator exactly;
3. decide whether the matched activation rate is population-defined, reference-estimated, or evaluation-estimated;
4. derive the joint evaluation and reference influence functions;
5. establish complete-replication paired bootstrap validity;
6. run runtime-only preflight;
7. lock the numerical protocol;
8. validate once and freeze the results.

No new large empirical model-fitting experiment is needed before D4.

18. Architecture decision log

Decision	Choice	Rationale
Number of papers	One unified top-tier paper; empirical v2 frozen as fallback	Avoid overlap and keep theory tied to a real phenomenon
First journal	Biometrika	Best match to conceptual novelty, foundational structure, and concise anomalous behavior
Main next theorem	D4 paired contrast	Directly closes the empirical/theoretical loop
D3 timing	After first complete draft	Prevent open-ended theory expansion
Main empirical example	Locked mixed-realistic confirmatory study	Strongest prespecified evidence
SUPPORT2 placement	Supplement, brief main-text reference	Preserves realism without overwhelming theory
TESS formula role	Characterized standardization	Avoid false novelty claim
Copula/extremal coefficient	Prior-art bridge	Locates fixed-family special case
D1 status	REVIEW, retained	Honest record of finite-sample basic-interval limitations
D2 status	PASS	Meets D0 core success criterion
Winner/tie/plus-one claim	Not yet covered by D2	Requires explicit bridge and scope statement